

Optimized Image Segmentation and Deep Learning Techniques for Accurate Plant Disease Detection

Dr M Ranjit Reddy¹, B Ramya², P Ayesha³, R Sai Sowmya⁴

¹ Professor, Dept of CSE, Srinivasa Ramanujan Institute of Technology, Anantapur, Andhra Pradesh

^{2,3,4} Students, Dept of CSE, Srinivasa Ramanujan Institute of Technology, Anantapur, Andhra Pradesh, India

Authors Email: ¹midderanjit@gmail.com, ²ramyabasava6@gmail.com, ³ayeshapalagiri265@gmail.com, ⁴rokkamsaisowmyasree@gmail.com

Abstract— Identifying plant diseases is a significant hurdle in maximizing agricultural productivity because it negatively impacts both yield and quality. Therefore, early and precise identification is crucial for effective disease management and monitoring. Traditional manual identification methods can be impractical and subjective, especially in large farming settings. On the other hand, current automated systems often analyze the entire leaf rather than focusing on the specific disease patches, which can lead to inaccuracies. To tackle these challenges, this project aims to implement a cutting-edge identification system that utilizes segmentation and various deep-learning algorithms to improve disease identification accuracy. The proposed system will first segment and preprocess the entire leaf to emphasize only the infected areas, and then apply different deep-learning algorithms, including a custom CNN, VGG19, and DenseNet201, using a dataset from Plant Village that features various crops with different diseases. The results from experiments conducted on this system confirm its effectiveness. Essentially, this indicates that the system has achieved efficient classification results by integrating deep learning techniques. Among all the methods tested, the DenseNet201 algorithm yielded the best performance, achieving an impressive accuracy rate of 98%. In addition to detecting and diagnosing plant diseases, the proposed system also outlines a comprehensive list of methods and techniques for managing these diseases.

Keywords— Plant Disease Classification, Image Segmentation, Deep Learning, Convolutional Neural Networks, Plant Village Dataset and Precision Agriculture

1. INTRODUCTION

Plant diseases are a significant threat to agricultural productivity across many parts of the globe. Each year, the serious loss in yield and quality due to these diseases results in substantial economic setbacks. Timely and accurate identification of plant diseases is crucial to prevent widespread damage, which in turn helps ensure food security and improves the quality of agricultural products. Traditionally, disease detection has relied on visual inspections by farmers or agricultural experts, but this method can be subjective, time-consuming, and often unreliable, especially when dealing with large farming areas. Recently, deep learning techniques have emerged and gained popularity; image-based methods for detecting plant diseases are now capturing a lot of attention. Many deep learning models have been introduced that can learn complex visual features

directly from raw images of plant leaves. However, most of these approaches classify images without adequately considering the infected areas. This highlights the need for an integrated approach that combines image segmentation with deep learning to achieve more accurate disease detection and provide valuable insights for agriculture. Accurately identifying the diseases affecting these plants is vital for boosting their yield and contributing to sustainable development worldwide. The urgency of this issue cannot be overstated; failing to address it will only allow these diseases to spread further, potentially leading to increased pest control measures, higher costs, and greater environmental damage. It's clear that expertise in this area, particularly within the agricultural sector, remains quite limited. Therefore, establishing a system that offers automated classifications and supports farmers is essential for tackling these challenges effectively.

2. LITERATURE REVIEW

In recent years, plant disease detection has gained a lot of attention, and for good reason—it's crucial for boosting agricultural productivity and ensuring we have enough food to go around. Initially, researchers leaned on traditional image processing techniques and machine learning methods to spot plant diseases. They would extract features like colour, texture, and shape from images of plant leaves, using classifiers such as Support Vector Machine (SVM) and K-Nearest Neighbour (KNN) to identify diseases. While these methods yielded decent results in controlled settings, their effectiveness often hinged on factors like lighting and image quality, which made them less reliable in the messy reality of agricultural fields. With the rise of deep learning, Convolutional Neural Networks (CNNs) have emerged as a go-to solution for detecting plant diseases.

These deep learning models can automatically learn intricate features from plant leaf images, eliminating the need for manual feature extraction. Numerous studies have utilized popular architectures like Alex Net, VGG, and Inception networks for classifying plant diseases, often using datasets such as the Plant Village dataset. The results have shown a marked improvement in accuracy compared to traditional machine learning techniques. However, a lot of existing deep learning methods tend to classify the entire leaf image without zeroing in on the infected areas. This can lead to background noise and irrelevant features muddying the classification results. Some research has primarily focused on boosting classification accuracy, overlooking practical considerations like disease prevention, usability, and actionable recommendations for farmers. Moreover, many previous systems have provided limited comparisons between different deep learning architectures, making it tough to pinpoint the most effective model for plant disease detection tasks.

To tackle these challenges, recent studies have been looking into the combination of image segmentation and deep learning techniques. Image segmentation is key in separating the

diseased areas of a plant leaf from the healthy sections and the background, which allows the deep learning model to concentrate on the important features. This method not only boosts classification accuracy but also minimizes the impact of background noise. Researchers are also comparing various deep learning architectures like CNN, VGG19, and Dense Net to see how effective they are in detecting plant diseases. Datasets like the Plant Village dataset are crucial for training and assessing plant disease detection models. This dataset includes thousands of labeled images of plant leaves from various crops and disease categories. These images offer rich visual details about disease symptoms, such as spots, discoloration, and texture changes, making it an excellent resource for training deep learning models for disease classification.

In summary, earlier research shows that deep learning techniques greatly enhance the accuracy of plant disease detection compared to traditional methods. However, by integrating segmentation techniques with deep learning models and conducting comparative analyses of different architectures, we can further improve performance and create more reliable disease diagnosis systems for precision agriculture.

3. EXISTING SYSTEM

Plant disease detection systems have traditionally leaned on classic image processing and machine learning techniques to spot issues in plant leaves. In these setups, images of leaves are taken and analyzed through various feature extraction methods, focusing on aspects like colour, texture, and shape. The features extracted are then classified using machine learning algorithms such as Support Vector Machine (SVM) and K-Nearest Neighbour (KNN) to figure out if a plant is healthy or suffering from disease. While these methods can achieve moderate accuracy, their effectiveness often hinges on the quality of the images and the surrounding environmental conditions.

With the rise of artificial intelligence, deep learning models, especially Convolutional Neural Networks (CNNs), have become popular for detecting plant diseases. Many current

systems utilize deep learning frameworks like Alex Net, VGG, and Inception networks to classify diseases based on leaf images. These models are trained on datasets like the Plant Village dataset, which boasts thousands of labeled images of plant leaves affected by various diseases. Deep learning techniques greatly enhance classification accuracy by automatically picking up on complex visual features from the images.

However, a lot of these systems tend to classify the whole leaf image without pinpointing the specific infected areas. This can lead to the inclusion of unnecessary background details and noise in the analysis, potentially lowering the accuracy of disease detection. Moreover, many existing models primarily aim to boost classification accuracy and often overlook providing disease prevention tips or decision-support tools for farmers.

4. PROPOSED SYSTEM

The proposed system is all about creating an automated plant disease detection tool that leverages image segmentation and deep learning techniques. Its primary aim is to accurately spot plant diseases by examining images of plant leaves. Rather than looking at the whole leaf image, this system zeroes in on just the infected parts of the leaf. This targeted approach minimizes background noise and enhances the accuracy of disease classification. By employing cutting-edge deep learning architectures, the system delivers reliable and efficient results for disease detection.

The process kicks off with gathering plant leaf images from the Plant Village dataset, which boasts thousands of labeled images across various crops and disease categories. Before these images are fed into the deep learning models, a few preprocessing steps are taken. This includes resizing the images, normalizing pixel values, and applying data augmentation techniques like rotation, flipping, and brightness adjustments. These preprocessing methods are crucial for boosting model performance and preventing overfitting during the training phase. Once the preprocessing is complete, the system moves on to image segmentation, which

separates the infected areas of the leaf from the healthy parts and the background. This segmentation allows the model to concentrate specifically on disease-related features such as spots, lesions, and discoloration patterns. By isolating the diseased sections of the leaf, the model can learn more pertinent features while minimizing the impact of irrelevant background information. This step plays a significant role in enhancing the overall accuracy of plant disease detection. After the initial preprocessing, the system dives into image segmentation, which is all about distinguishing the infected parts of the leaf from the healthy areas and the background. This process allows the model to zero in on disease-related features like spots, lesions, and discoloration patterns. By isolating the sick parts of the leaf, the model can pick up on more relevant characteristics while filtering out any distracting background noise. This crucial step really boosts the overall accuracy of detecting plant diseases.

The segmented images are then fed into deep learning models for classifying the diseases. In this setup, three different architectures come into play: a Custom Convolutional Neural Network (CNN), VGG19, and DenseNet201. These models automatically pull out key features from the images and sort them into various disease categories. Out of these, DenseNet201 stands out with its impressive performance, thanks to its dense connectivity that enhances feature reuse and improves gradient flow during training. Experimental results indicate that DenseNet201 achieves a higher classification accuracy compared to the other models.

In the end, the proposed system does more than just spot plant diseases; it also offers recommendations for prevention and control. Based on the identified disease, the system suggests potential treatment options and preventive strategies for farmers. This makes the system not only practical but also incredibly useful in real agricultural settings. By combining image segmentation, deep learning classification, and advisory support, this system emerges as a powerful solution for smart plant disease detection and management.

5. SYSTEM ARCHITECTURE

The system architecture for the proposed plant disease detection model is crafted to efficiently handle plant leaf images and accurately identify diseases using deep learning techniques. It's structured in several stages: image input, preprocessing, image segmentation, deep learning classification, and result output. Each of these stages plays a crucial role in enhancing the accuracy and reliability of the disease detection process.

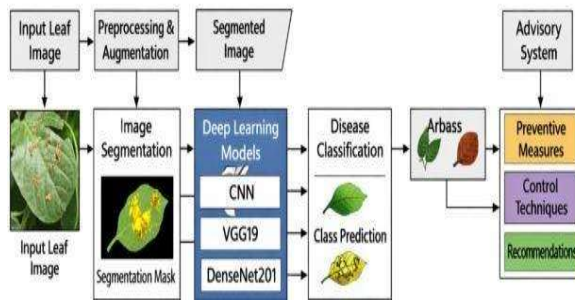


Fig.1 System Architecture

Let's start with the first stage, which is all about image acquisition. Here, we gather plant leaf images from the Plant Village dataset, a treasure trove of labeled images showcasing various plant diseases alongside healthy leaf conditions. These images are the foundation for training and testing our deep learning models. The dataset offers clear, high-resolution images of plant leaves, enabling the model to effectively learn the different patterns associated with diseases.

Once we have our images, we move on to the preprocessing stage. This is where we resize the input images to a standard dimension that fits the needs of our deep learning models. We also normalize the pixel values to ensure consistency in intensity across the images. To further enrich our dataset, we apply data augmentation techniques like rotation, flipping, zooming, and adjusting brightness. This not only diversifies our dataset but also boosts the model's ability to adapt to new images.

Next up is image segmentation, a vital step where we isolate the infected areas of the leaf from the healthy parts and the background. This process emphasizes the diseased sections, allowing the model to concentrate on key

features like spots, lesions, and discoloration patterns. By reducing background noise, we enhance feature extraction during the classification phase, making the whole process more effective.

After segmentation, the processed images move on to the deep learning classification phase. Here, various deep learning models like Custom CNN, VGG19, and DenseNet201 come into play to analyze the segmented leaf images. These models are designed to automatically pull out intricate features from the images using convolution layers and sort them into different categories of plant diseases. Among these options, DenseNet201 stands out for its ability to efficiently reuse features and enhance gradient flow.

In the final output stage, the system delivers the predicted disease class along with helpful recommendations for prevention and control. To assess how well the models are performing, metrics such as accuracy, precision, recall, and F1-score are utilized. This entire architecture is crafted to ensure effective disease detection, boost classification accuracy, and provide practical assistance to farmers in managing plant diseases.

6. METHODS AND MATERIALS

A. Data Collection and Preprocessing:

The system we're proposing relies on the Plant Village dataset to train and assess our plant disease detection model. This dataset is quite extensive, featuring around 54,000 labeled RGB images of plant leaves that showcase various crop species and disease types, including both healthy and unhealthy leaves. These images offer clear visual cues about disease symptoms, like spots, discoloration, and changes in texture. Before we dive into model training, we apply several preprocessing steps to enhance the quality and consistency of the data. We resize the images to a standard resolution that works well with deep learning models, and we normalize the pixel values to ensure uniform intensity levels across all images. Additionally, we employ data augmentation techniques—such as rotation, flipping, zooming, and adjusting brightness—to boost the dataset's diversity and

help prevent overfitting. These preprocessing measures are crucial for enhancing the model's ability to generalize and accurately classify diseases.

B. Image Segmentation:

Image segmentation plays a crucial role in pinpointing and isolating the infected areas on plant leaves. Rather than examining the whole leaf image, this process effectively distinguishes the diseased sections from the healthy parts and the background. By doing so, it enables deep learning models to concentrate on the key features of the disease, like lesions, spots, and colour changes. By filtering out unnecessary background details, segmentation enhances the feature extraction process and boosts the accuracy of disease classification.

C. Deep Learning Models:

After the segmentation process, the images are sent off to the classification stage, where deep learning models come into play to identify plant diseases. In this system, we've got three different models at work: a Custom **Convolutional Neural Network (CNN)**, **VGG19**, and **DenseNet201**. These models are pretty impressive—they automatically pull out complex visual features from the leaf images using convolution and pooling layers. For VGG19 and DenseNet201, we use transfer learning with pre-trained ImageNet weights, which really boosts both training efficiency and classification performance. Out of all these models, DenseNet201 stands out as the top performer, thanks to its dense connectivity that allows for efficient feature reuse and better gradient flow during training.

D. Model Training and Evaluation:

During training, we optimize the models with the Adam optimizer, which does a great job of fine-tuning the learning rates. For identifying different plant diseases, we rely on the categorical cross-entropy loss function, especially when dealing with multiple classes. The models go through several epochs, using carefully chosen batch sizes to keep the training balanced and effective. To see how well the system performs, we look at metrics like accuracy, precision, recall, and F1-score. Our experiments reveal that the DenseNet201 model

stands out, achieving an impressive accuracy of about 98%. This really highlights how effective the segmentation-assisted deep learning approach is for detecting plant diseases.

E. User Module:

The User Module is designed for farmers and users to engage with the plant disease detection system. Once registered and logged in, users can easily access the system's services. After logging in, they can upload images of plant leaves right through the system's interface. The deep learning model then processes these images to determine if the plant is healthy or suffering from a disease. Following this analysis, the system provides information on the predicted disease along with suggestions for prevention and treatment. Additionally, users have the option to review their previously uploaded images and analysis results for future reference.

F. Admin Module:

The Admin Module plays a crucial role in overseeing and managing the entire system. It gives the admin the ability to keep an eye on all user activities and dive into the results produced by the system. With this module, the admin can check out registered users, handle user accounts, and keep track of uploaded images along with disease detection outcomes. The admin also has the power to update the dataset, introduce new disease information, and take care of the system database. On top of that, the admin can create reports and assess system performance to make sure everything runs smoothly and delivers precise disease detection results.

7. RESULTS

The experimental results show that the system we've proposed is both accurate and reliable when it comes to detecting plant diseases. By combining image segmentation with deep learning models, we've significantly improved how features are extracted, which in turn boosts classification accuracy. Plus, the system offers practical recommendations for preventing and controlling diseases, making it a valuable tool for real-world agricultural use. These results highlight the strong potential of our framework to assist farmers and agricultural experts in

efficiently identifying plant diseases and ultimately enhancing crop productivity.

To effectively showcase how the AgriScan plant disease detection system works and performs, this section includes several interface screenshots. These images highlight key components of the system, such as user authentication pages, plant image analysis modules, dashboards, and the disease detection results produced by the deep learning model. By displaying these interfaces, we can better understand how the system functions in real-time and how users engage with the platform to identify plant diseases.

The system employs image preprocessing, segmentation, and advanced deep learning classification models like CNN, VGG19, and DenseNet201 to accurately analyze plant leaf images and detect diseases. The results from the model are presented through intuitive interfaces, allowing both farmers and administrators to keep an eye on crop health and system performance. The following figures illustrate the Agri Scan system's working interfaces, showcasing its capabilities to upload plant images, analyze diseases, display scan statistics, and deliver detailed diagnostic results.

Model / Approach	Accuracy (%)
Traditional CNN	89.4
VGG16	91.8
VGG19	93.2
ResNet50	94.6
DenseNet201	95.1
CNN (with augmentation)	96.0
Segmentation+ DenseNet201	98.2

Figure 1: Comparison of Existing Models vs Proposed Method

These interface screenshots also emphasize the functionalities of both the user and admin modules. The user module allows farmers to upload leaf images and access disease detection results, while the admin module offers analytical dashboards, tracks scan history, and oversees overall system performance. By presenting these visual interfaces, we can clearly grasp the effectiveness and usability of the proposed system.



Figure 2: Landing Page

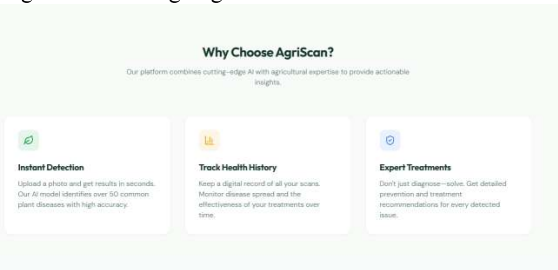


Figure 3: Platform Features

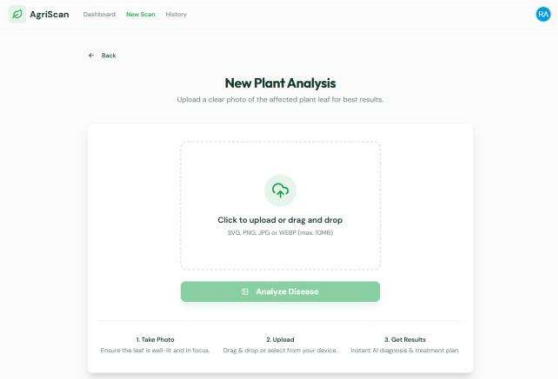


Figure 4: Upload Plant Image for Analysis

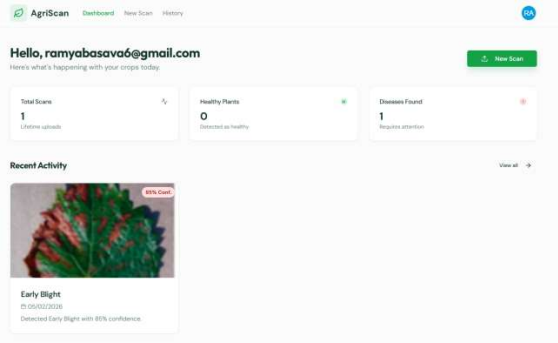


Figure 5: User Dashboard

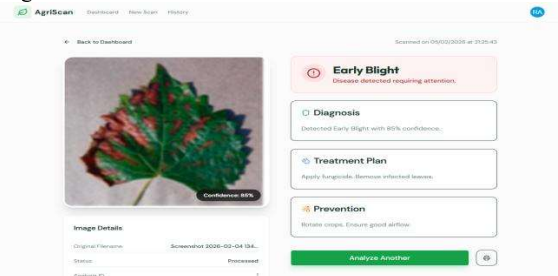


Figure 6: Disease Detection Result

This interface displays the final disease detection result generated by the deep learning model. It shows the

identified disease type along with confidence levels and possible recommendations for treatment or prevention.

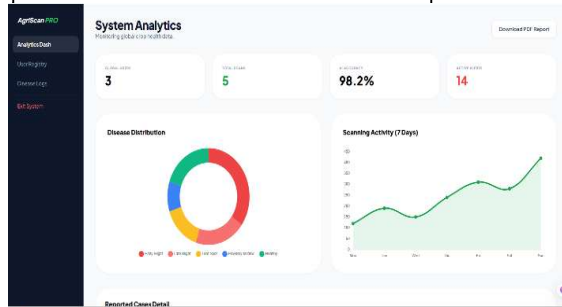


Figure 7: Admin Dashboard

The admin dashboard provides an overview of system analytics including user statistics, total scans, and disease detection trends. It helps administrators monitor platform usage and system performance.

8. DISCUSSION

The findings indicate that the system is quite effective at spotting plant diseases by utilizing image segmentation and deep learning models. By concentrating on the infected areas of leaf images, the system enhances feature extraction while minimizing background noise, which ultimately boosts the accuracy of disease classification.

Among the models tested, DenseNet201 stood out with an impressive accuracy of around 98%, surpassing both CNN and VGG19 models. This highlights how advanced deep learning architectures, when paired with the right preprocessing techniques, can greatly enhance the performance of plant disease detection.

Additionally, the system offers practical advantages through user and admin dashboards that make it easy to upload images, analyze diseases, and keep track of scan statistics. While the system performs admirably, there's room for improvement, such as training with more real-field images to further refine accuracy across various environmental conditions.

9. CONCLUSION AND FUTURE WORK

The system showcases a smart way to detect plant diseases by blending image segmentation with deep learning models. It zeroes in on the infected areas of plant leaves and utilizes models like CNN, VGG19, and DenseNet201 to boost the accuracy of disease classification. The experimental findings reveal that DenseNet201

stands out with an impressive accuracy of about 98%, demonstrating that using segmentation alongside deep learning can really elevate the performance of plant disease detection.

The system also features user-friendly interfaces that allow farmers to easily upload images of their plants and get quick disease diagnosis results. Meanwhile, the admin dashboard is designed to keep track of system activities and analyze scan data. These functionalities make the platform not just theoretical but practical for real-world agricultural use, ultimately aiding in better crop management.

Looking ahead, there's room for improvement by adding more real-world field images to enhance the model's robustness across various environmental conditions. It would also be beneficial to include additional plant species and disease categories to broaden the system's capabilities. Plus, integrating the system into mobile apps or real-time monitoring tools could empower farmers to swiftly identify plant diseases and take preventive measures to safeguard their crops.

10. REFERENCE

1. Shahbazi, F., Shahbazi, S., Nadimi, M., & Paliwal, J. (2025). Losses in agricultural produce: A review of causes and solutions, with a specific focus on grain crops. *Journal of Stored Products Research*, 111, 102547.
2. Rao, K. S., Haran, R. H., & Rajpoot, V. S. (2022). Value addition: A novel strategy for quality enhancement of medicinal and aromatic plants. *Journal of Applied Research on Medicinal and Aromatic Plants*, 31, 100415.
3. Panchal, A. V., Patel, S. C., Bagyalakshmi, K., Kumar, P., Khan, I. R., & Soni, M. (2023). Image-based plant diseases detection using deep learning. *Materials Today: Proceedings*, 80, 3500-3506.
4. Upadhyay, A., Chandel, N. S., Singh, K. P., Chakraborty, S. K., Nandede, B. M., Kumar, M., ... & Elbeltagi, A. (2025). Deep learning and computer vision in plant disease detection: a comprehensive review of techniques, models, and trends in precision agriculture. *Artificial Intelligence Review*, 58(3), 92.
5. Costa, F., Frecassetti, S., Rossini, M., & PortioliStaudacher, A. (2023). Industry 4.0 digital technologies enhancing sustainability: Applications and barriers from the agricultural industry in an emerging economy. *Journal of Cleaner Production*, 408, 137208.